

Control Group Consortium

Group 13: Lauren K. Paulina S.



Scenario *Overview*

Control Group Consortium

- Collaborative sharing research facilities, equipment, and staff
- Facilities: wet labs, imaging suites, clean rooms, fabrication spaces, etc.
- Researchers submit requests to schedule experiments
- Resources are limited. Staff help sort conflicts and propose alternatives
- Experiments are dynamic: they extend, shrink, reschedule, or cancel
- Staff support setup, operation, and maintenance
- Usage generates charges billed to grants, departments, or external sponsors



Content *overview*

1

ERD

2

Schema
Design

3

Queries

ERD *Sections*

Charge



Experiment schedule
and participating entities



Facility



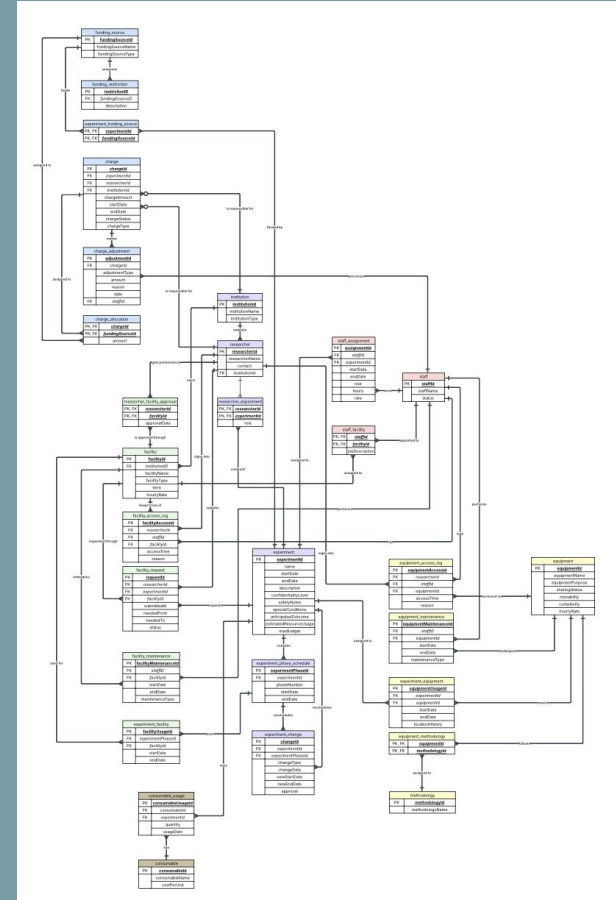
Equipment



Consumable



Staff



ERD

Sections

Facility

- facility
- researcher_facility_approval
- facility_access_log
- facility_request
- facility_maintenance
- experiment_facility

Equipment

- equipment
- equipment_access_log
- equipment_maintenance
- experiment_equipment
- equipment_methodology
- methodology

Experiment schedule and participating entities

- experiment
- experiment_phase_schedule
- experiment_change
- institution
- researcher
- researcher_experiment

Staff

- staff
- staff_assignment
- staff_facility

Consumable

- consumable
- consumable_usage

Charge

- charge
- charge_adjustment
- charge_allocation
- experiment_funding_source
- funding_restriction
- funding_source

Schema *Design* and *Implementation*

- Using SQLite database
- In controlgroup.db, we created relevant tables by running create.sql
- To populate the database, we ran insert.sql to fill in data

```
CREATE TABLE researcher (
    researcherId INTEGER PRIMARY KEY,
    researcherName TEXT NOT NULL,
    contact TEXT NOT NULL,
    institutionId INTEGER NOT NULL,
    FOREIGN KEY (institutionId) REFERENCES institution(institutionId)
);

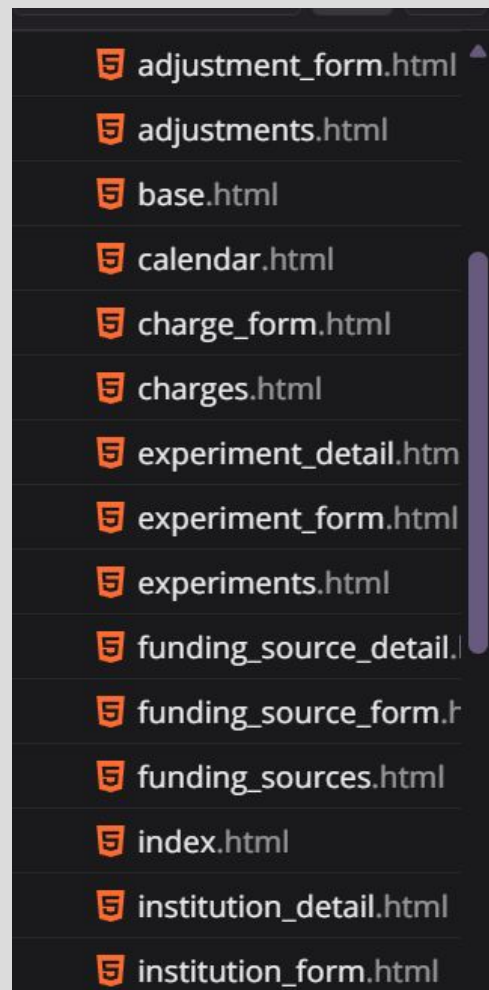
CREATE TABLE experiment (
    experimentId INTEGER PRIMARY KEY,
    name TEXT NOT NULL,
    startDate TEXT NOT NULL,
    endDate TEXT NOT NULL,
    description TEXT NOT NULL
);
```

```
INSERT INTO charge VALUES
(1,1,5000,'2025-01-15','2025-02-15','Equipment'),
(2,2,8000,'2025-02-10','2025-03-10','Software'),
(3,3,6000,'2025-02-20','2025-03-25','Lab'),
(4,4,7000,'2025-03-10','2025-04-10','Staff'),
(5,5,6500,'2025-02-25','2025-03-30','Materials'),
(6,6,9000,'2025-03-20','2025-04-25','Robotics'),
(7,1,3000,'2025-03-01','2025-03-31','Maintenance'),
(8,2,4000,'2025-03-15','2025-04-15','Cloud');

INSERT INTO charge_adjustment VALUES
(1,1,'Discount',-500,'2025-02-01'),
(2,2,'Extra Cost',1000,'2025-03-05'),
(3,3,'Correction',-300,'2025-03-15'),
(4,4,'Bonus Funding',1200,'2025-04-01'),
(5,5,'Discount',-200,'2025-03-10'),
(6,6,'Extra Cost',1500,'2025-04-05'),
(7,7,'Correction',-100,'2025-03-20'),
(8,8,'Bonus Funding',800,'2025-04-10');
```

EdStem Repo

- app.py + many .html files
 - For each “feature” page, e.g. a form to add a new researcher, institution, etc.
- Converted from original CLI to full web application, as well as adding many more features
 - E.g. Search feature



Site + Highlights

[CSCI-UA 60 001 – Ed Workspaces](#)

- Quick actions to edit database
- Quick navigation with tabs, or searching in the search bar
- Visually pleasing compared to CLI
 - Calendar
 - Data display
- Experiment calendar displaying anything upcoming

Thank you

for listening!